Hasse Diagram
Chain and Antichain
Function

Theorem

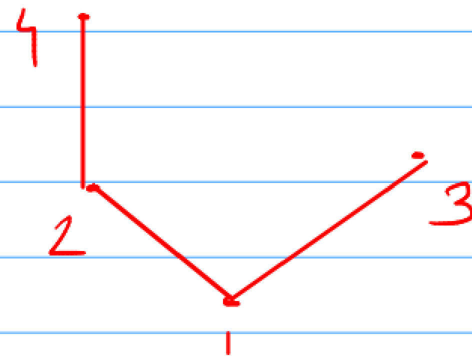
- **Let (P, ≤) be a partially ordered set. Suppose the length of the longest chains in P is n, then the elements in P can be partitioned into n disjoint antichains.**

Example

Ex:

$P = \{1, 2, 3, 4\}$

$R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$



length of longest chain $(n) = 3$

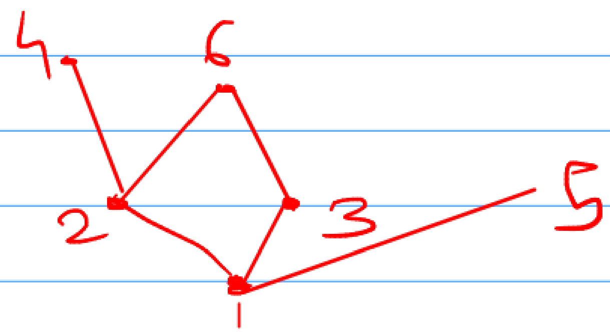
3 disjoint antichains

$\{3,4\}, \{1\}, \{2\}$

Example

Ex:

$P = \{1, 2, 3, 4, 5, 6\}$



length of longest chain $n = 3$

3 disjoint antichains

$= \{1\}, \{2, 3\}, \{4, 5, 6\}$

Proof of theorem by induction method

- **Let (P, ≤) be a partially ordered set. Suppose the length of the longest chains in P is n, then the elements in P can be partitioned into n disjoint antichains.**

Proof

- Basis of Induction : For $n=1$, no two elements in P are related. Clearly, they constitute an antichain.

$N=1$ $a^{\bullet}$ $\dot{b}$ $\dot{c}$ $\dot{d}$

length of longest chain $= 1$

Partition $P = \{a, b, c, d\}$

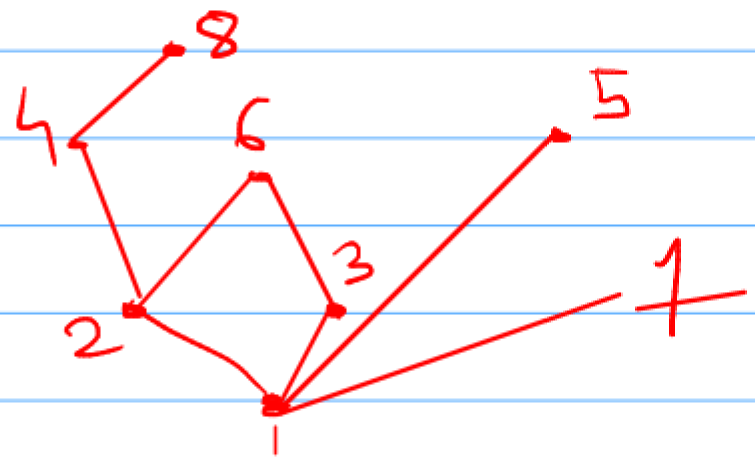
~ Hypothesis:

We assume that the theorem holds when the length of the longest chains in a partially ordered set is $n-1$.

~ Induction Step

~ Let P be a partially ordered set with longest chain length $= n$

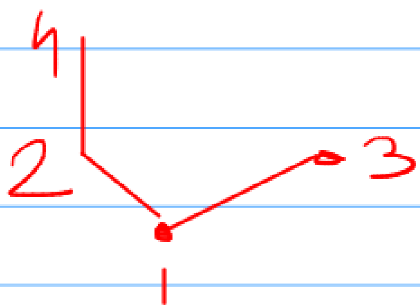
$P = \{1, 2, 3, 4, 5, 6, 7, 8\}$



~ Length of longest Chain $= n$

~ Let m be Set of maximal Elements
$m = \{5, 6, 7, 8\}$

~ Consider Set $P - M = \{1, 2, 3, 4\}$



~ Length of longest chain in $P - M$ is $(n-1) = 3$.

~ Length of longest chain in P-M is $(n-1) = 3$.

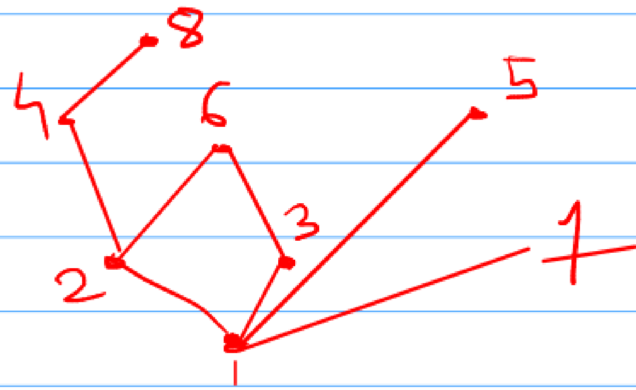
~ As per hypothesis, theorem holds for set $P - M$.

~ Now, $P = (P - M) \cup M$

~ So, P can be partitioned into $(n-1) + 1 = n$ disjoint Antichains ■

Corollary:

- Let $(P, \leq)$ be a Partially ordered set consisting of $mn+1$ elements, either there is an antichain consisting of $m+1$ elements or there is a chain of length $n+1$ in P.



length of longest chain = 4

$\{1, 2, 3, 4, 5, 6, 7, 8\}$

$mn+1 = 8$

$mn = 7$

$m=1, n=7$

There is antichain Consisting of 2 Elements

$\{3, 4\}$

Functions

Functions

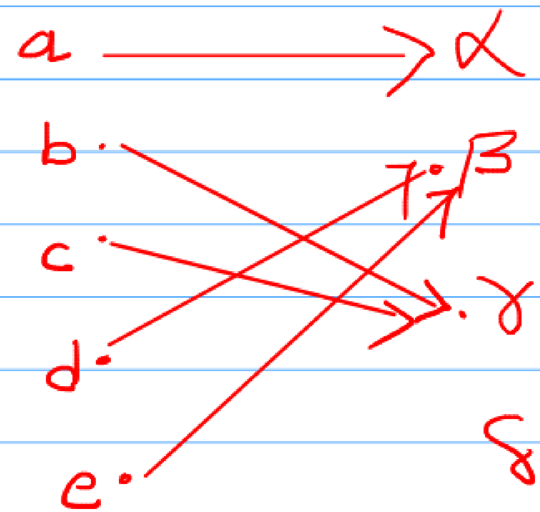
- A binary relation R from A to B is said to be a function if for every element a in A, there is a unique element b in B, so that (a,b) is in R.
- Notation R(a) = b

Example

– Let

$A = \{a, b, c, d, e\}$

$B = \{\alpha, \beta, \gamma, \delta\}$



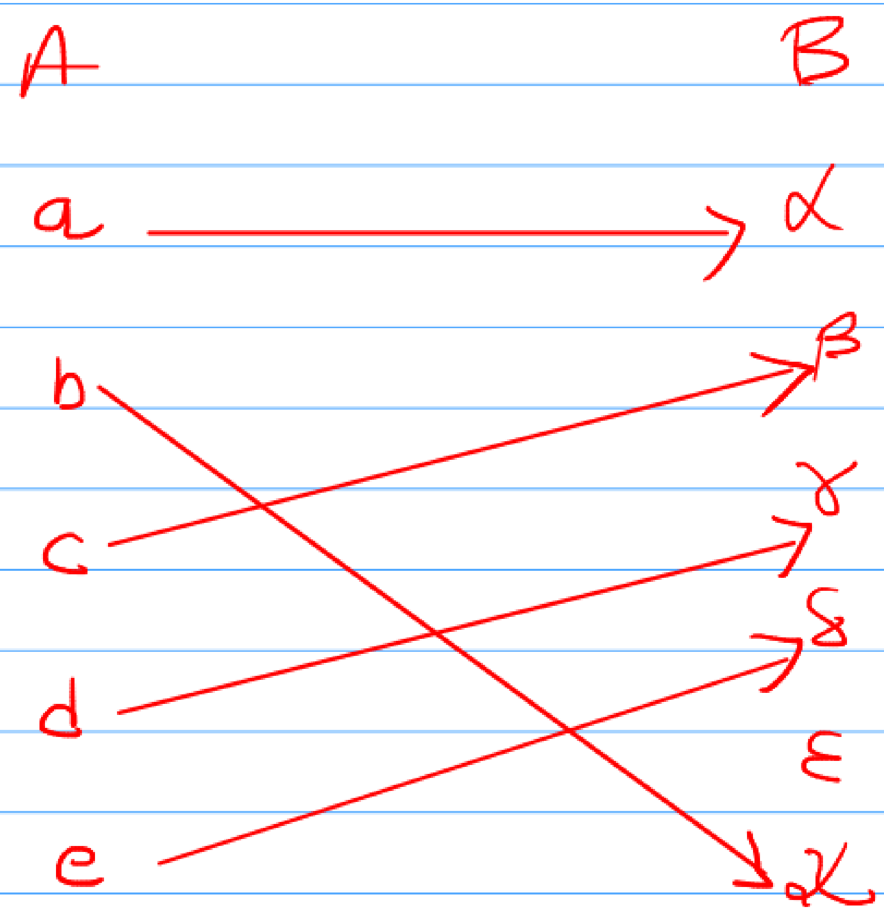
- A **General Function** points from each member of "A" to a member of "B".
- It **never** has one "A" pointing to more than one "B", so **one-to-many is not OK** in a function (so something like "f(x) = 7 ***or*** 9" is not allowed)
- But more than one "A" can point to the same "B" (**many-to-one is OK**)

One-to-One (**Injective**)

- A function from A to B is said to be a one-to-one function if no two elements of A have the same image.

Example

- Let A be set of workers and B be set of Jobs. One to one function from A to B is assignment of function in such a way that no two workers have the same job.

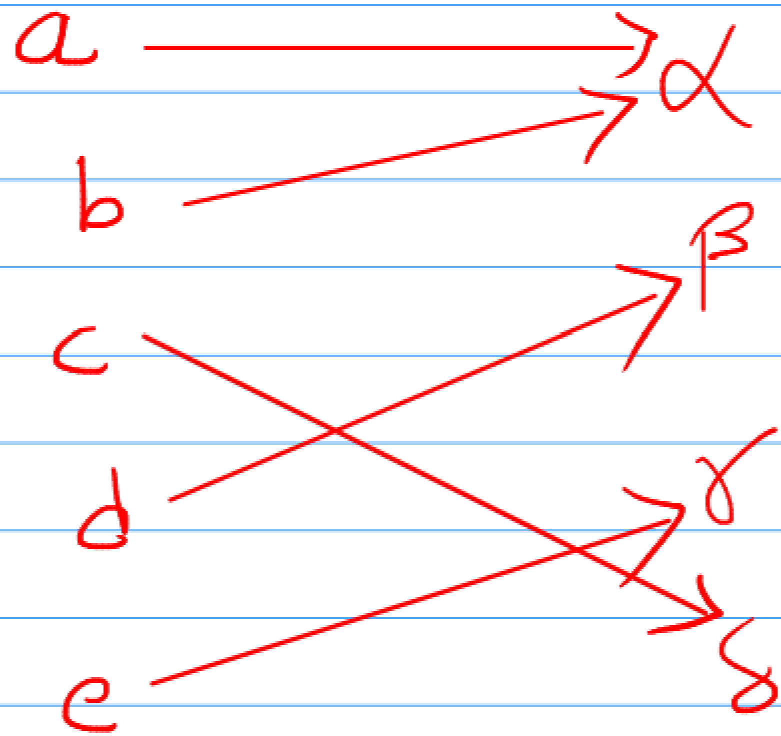

A
B
a
α
b
β
γ
c
d
δ
ε
e

Onto function (surjective)

- A function from A to B is said to be an onto function if every element of B is the image of one or more elements of A.

Example

- Let A be set of workers and B be set of Jobs. onto function from A to B is assignment of function in such a way that every job has at least one worker assigned to it.


a
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One-to-One Onto function (bijective)

- A function from A to B is said to be bijective if it is both an onto and a one to one function.

Ex. To show a function is one-to-one assume $f(\alpha) = f(\beta)$ and show that $\alpha = \beta$.

1. $f: R \to R$, $f(x) = 2x + 3$

let $f(\alpha) = f(\beta)$

$$2\alpha + 3 = 2\beta + 3$$

$$2\alpha + 3 - 2\beta - 3 = 0$$

$$\alpha - \beta = 0$$

$$\boxed{\alpha = \beta}$$

So, f is one to one

$f: R \to R, \quad f(x) = x^2.$

Let $f(\alpha) = f(\beta)$

$$\alpha^2 = \beta^2$$

$$\alpha^2 - \beta^2 = 0$$

$$(\alpha - \beta)(\alpha + \beta) = 0$$

$\alpha = \beta$ or $\alpha = -\beta$

So, f is not one to one

Onto function

- f : x->y. For every y in the co domain, there exists some x in the domain such that f(x)=y.

Ex. $f: N \to R$, $f(x) = 2x+3$

Let, $f(x) = y$

$\therefore 2x+3 = y$

$\therefore x = \frac{y-3}{2}$

Since $y \in R$, $\frac{y-3}{2} \in R$, But $x \in N$

So, for every y, there will not be corresponding x.

Ex. if $y = 4$, then x should be $\frac{1}{2}$ which is not possible

Example

Ex A function f is defined $f : R^+ \to N$

$f(x) = \lfloor x \rfloor$ (the largest int $\leq x$)

Is f, one to one, onto or both?